

# AFRILANG Stdlib

## std/c/csvpivot

**Français.** Bibliothèque AFRILANG 'std/c/csvpivot' (niveau complex, catégorie complex). Elle expose 8 fonction(s) : pivoSum, pivoMean, pivoMin, pivoMax, pivoVar, pivoStd, pivoNorm, filterGt.

```
'import "std/c/csvpivot"' · 'use csvpivot'
```

```
- 'pivoSum(v list number) → number' - 'pivoMean(v list number) → number' - 'pivoMin(v list number) → number' - 'pivoMax(v list number) → number' - 'pivoVar(v list number) → number' - 'pivoStd(v list number) → number' - 'pivoNorm(v list number) → list number' - 'filterGt(col list number, threshold number) → list number'
```

**English.** AFRILANG library 'std/c/csvpivot' (tier complex, category complex). Exports 8 function(s): pivoSum, pivoMean, pivoMin, pivoMax, pivoVar, pivoStd, pivoNorm, filterGt.

```
'import "std/c/csvpivot"' · 'use csvpivot'
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- 'pivoSum(v list number) → number' - 'pivoMean(v list number) → number' - 'pivoMin(v list number) → number' - 'pivoMax(v list number) → number' - 'pivoVar(v list number) → number' - 'pivoStd(v list number) → number' - 'pivoNorm(v list number) → list number' - 'filterGt(col list number, threshold number) → list number'
```

|                       |                      |               |
|-----------------------|----------------------|---------------|
| Catégorie / Category  | Modules complex (c/) |               |
| Niveau / Tier         | complex              | June 16, 2026 |
| Fonctions / Functions | 8                    |               |

## Contents

## Français

### 1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/csvpivot' (niveau complex, catégorie complex). Elle expose 8 fonction(s) : pivoSum, pivoMean, pivoMin, pivoMax, pivoVar, pivoStd, pivoNorm, filterGt.

```
'import "std/c/csvpivot"' · 'use csvpivot'
```

```
- `pivoSum(v list number) → number`
- `pivoMean(v list number) → number`
- `pivoMin(v list number) → number`
- `pivoMax(v list number) → number`
- `pivoVar(v list number) → number`
- `pivoStd(v list number) → number`
- `pivoNorm(v list number) → list number`
- `filterGt(col list number, threshold number) → list number`
```

### 2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/csvpivot"
use csvpivot
```

Niveau : complex · Catégorie : Modules complex (c/)  
Domaines avancés – graphes, NLP, réseaux complexes.

### 3. Référence API

- pivoSum(v list number) → number•  
pivoMean(v list number) → number•  
pivoMin(v list number) → number•  
pivoMax(v list number) → number•  
pivoVar(v list number) → number•  
pivoStd(v list number) → number•  
pivoNorm(v list number) → list•  
filterGt(col list number, threshold number) → list

### 4. Exemples d'utilisation

```
import "std/c/csvpivot"
use csvpivot
```

```
create result = pivoSum(/* args */)
say result
```

```
create result = pivoMean(/* args */)
say result
```

```
create result = pivoMin(/* args */)
say result
```

```
create result = pivoMax(/* args */)
say result
```

```
create result = pivoVar(/* args */)
say result
```

```
create result = pivoStd(/* args */)
say result
```

## 5. Bonnes pratiques

- Préférez ``import "std/c/csvpivot"`` en tête de fichier.
- Lancez ``afrilang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

## 6. Annexe — source

module csvpivot

```
function pivoSum(v list number) returns number
end
```

```
function pivoMean(v list number) returns number
end
```

```
function pivoMin(v list number) returns number
end
```

```
function pivoMax(v list number) returns number
end
```

```
function pivoVar(v list number) returns number
end
```

```
function pivoStd(v list number) returns number
end
```

```
function pivoNorm(v list number) returns list number
end
```

```
function filterGt(col list number, threshold number) returns list number
end
end
```

## English

### 1. Overview

AFRILANG library 'std/c/csvpivot' (tier complex, category complex). Exports 8 function(s): pivoSum, pivoMean, pivoMin, pivoMax, pivoVar, pivoStd, pivoNorm, filterGt.

```
'import "std/c/csvpivot"' · 'use csvpivot'
```

```
- `pivoSum(v list number) → number`
- `pivoMean(v list number) → number`
- `pivoMin(v list number) → number`
- `pivoMax(v list number) → number`
- `pivoVar(v list number) → number`
- `pivoStd(v list number) → number`
- `pivoNorm(v list number) → list number`
- `filterGt(col list number, threshold number) → list number`
```

### 2. Installation & import

Add the module to your program:

```
import "std/c/csvpivot"
use csvpivot
```

Tier: complex · Category: Complex modules (c/)  
Advanced domains – graphs, NLP, complex networks.

### 3. API reference

- - pivoSum(v list number) → number•
  - pivoMean(v list number) → number•
  - pivoMin(v list number) → number•
  - pivoMax(v list number) → number•
  - pivoVar(v list number) → number•
  - pivoStd(v list number) → number•
  - pivoNorm(v list number) → list•
  - filterGt(col list number, threshold number) → list

### 4. Usage examples

```
import "std/c/csvpivot"
use csvpivot
```

```
create result = pivoSum(/* args */)
say result
```

```
create result = pivoMean(/* args */)
say result
```

```
create result = pivoMin(/* args */)
say result
```

```
create result = pivoMax(/* args */)
say result
```

```
create result = pivoVar(/* args */)
say result
```

```
create result = pivoStd(/* args */)
say result
```

## 5. Best practices

- Prefer ``import "std/c/csvpivot"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

## 6. Appendix — source

module csvpivot

```
function pivoSum(v list number) returns number
end
```

```
function pivoMean(v list number) returns number
end
```

```
function pivoMin(v list number) returns number
end
```

```
function pivoMax(v list number) returns number
end
```

```
function pivoVar(v list number) returns number
end
```

```
function pivoStd(v list number) returns number
end
```

```
function pivoNorm(v list number) returns list number
end
```

```
function filterGt(col list number, threshold number) returns list number
end
end
```

## Référence rapide / Quick reference

- Import : `import "std/c/csvpivot"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/csvpivot/>

## Source (extrait)

```
module csvpivot

function pivoSum(v list number) returns number
end

function pivoMean(v list number) returns number
end

function pivoMin(v list number) returns number
end

function pivoMax(v list number) returns number
end

function pivoVar(v list number) returns number
end

function pivoStd(v list number) returns number
end

function pivoNorm(v list number) returns list number
end

function filterGt(col list number, threshold number) returns list number
end
end
```